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Research Paper

Machine learning study of bubble nucleation and dynamics on a boiling surface based on lattice Boltzmann method

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ABSTRACT

Pool boiling is a common and efficient two-phase heat transfer process, which has wide and important applications in the fields of heat exchange, temperature control, and thermal management in modern industries. Bubble dynamics play a crucial role in pool boiling processes. These dynamic processes include bubble nucleation, growth, departure from the heated surface, and complex interactions between bubbles. A well-designed surface engineering approach can optimize these dynamic processes, thereby significantly enhancing the overall heat transfer efficiency. But how the surface parameters affect phase change and heat transfer behavior by influencing bubble dynamics has not been elucidated. In light of this, this study uses a method combining the Lattice Boltzmann method (LBM) and machine learning (ML) method to investigate the effects of surface parameters on the heat transfer process during bubble nucleation. A numerical simulation study of pool boiling on a flat surface is first performed. The simulation framework is based on the pseudopotential method for two-phase flow and the double distribution function method for thermal simulation. This part examine how surface density affects wettability and wetting modes, while also considering the impact of surface roughness. Typical bubble nucleation processes of pool boiling are obtained. A data mining process for contact line dynamics parameters based on ML method and computer vision is further applied to obtain the contact line dynamics data related to the speed, acceleration, and contact angle of bubble three-phase contact lines (TCLs) during the pool boiling process. The results demonstrate the effects of different surface characteristics on bubble nucleation, surface wetting, heat transfer behavior, and contact line dynamics in pool boiling. Surface roughness and wettability can both enhance the heat transfer behavior through the influence of the bubble TCLs.

1. Introduction

Pool boiling is a thermal process where a liquid undergoes phase transformation to gas through heating from a surface. Compared with single-phase heat transfer processes, the violent disturbances caused by bubble generation, coalescence, and departure in pool boiling greatly enhances the heat transfer coefficient (HTC) [1]. Pool boiling technology is widely employed in power generation, chemical processing, refrigeration systems, and food processing industries. A deep understanding and optimization of this process are crucial for technological advancement and industrial applications. However, owing to the inherent multiscale complexity of during bubble nucleation, there are significant challenges in optimizing the thermal efficiency of pool boiling. How to improve heat transfer efficiency of nucleate boiling on real-world surfaces with complex physicochemical properties has become a highly focused research topic.

Efficiency of pool boiling is influenced by various factors including the properties of the liquid, surface superheat, and environmental conditions. The characteristics of the heating surface play a pivotal role in affecting the efficiency of pool boiling. Surface properties, including surface roughness [2], wettability [3], and microstructural features [4], significantly influence the boiling behavior of the liquid. Specifically, surface roughness can increase the contact area of the boiling surface, allowing more liquid molecules to interact with the hot surface, thereby promoting the nucleation of more bubbles. Modifying the surface material by enhancing its wettability can also effectively control the nucleation and growth of bubbles on the surface, thus impacting the efficiency and stability of heat transfer. Many efforts have been

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Nomenclature

Alphabet

$\boldsymbol{e}_i$	lattice velocity vector
$\boldsymbol{u}$	velocity
Δt	simulation timestep
ΔT^*	surface superheat
a^*_{TCL}	dimensionless acceleration of TCL
c_s	lattice sound speed
Ca	capillary number
f_i	particle distribution function
g_i	temperature particle distribution function
h	heat transfer coefficient
$N_{\mathrm{b,max}}$	maximum bubble number
N_{b}	bubble number
q	heat flux
Ra	surface arithmetic average roughness
T	temperature
t	time
T_{c}	critical temperature
T_{w}	surface temperature
u^*_{TCL}	dimensionless speed of TCL

Greek symbols

μ	dynamic viscosity
Ω_i	collision operator
$\overline{\phi}$	time-averaged void fraction
ϕ	void fraction
ψ	pseudopotential function
ρ	density
σ	surface tension
τ	relaxation time
θ	droplet contact angle
θ^*	bubble contact angle
ξ	surface density

Subscripts

avg	average
b	bubble
c	critical
d	droplet
f	fluid
s	saturation
w	surface
i	i-th lattice direction

made to adopt surface engineering methods to improve heat transfer performance during pool boiling. Liang et al. [5] reviewed various surface modification techniques for enhancing pool boiling, including macroscale, microscale, nanoscale, multiscale, and hybrid-wettability methods. This work highlighted the challenges of implementing these techniques in temperature-sensitive applications like cooling electronic and power devices. Sajjad et al. [6] reviewed how surface engineering improves the boiling process of these liquids for better heat transfer with respect to dielectric liquids. The study summarized and classified the HTC and critical heat flux (CHF) of various engineered surfaces under different conditions, and certain surface modification techniques that may reduce performance and evaluate existing prediction methods, such as empirical models and artificial intelligence (AI) based models [7]. Shen et al. [8] enhanced low-pressure water boiling by using a dimpled, PTFE-coated copper surface, improving heat transfer by up to 60% at 9.5 kPa. Results indicated that gas trapping in hydrophobic cavities aids bubble formation, though effectiveness decreases below this pressure.

Recent advancements in high-performance computing devices, microfluidic devices, and microelectromechanical systems have sparked increased interest in studying pool boiling phenomena within small, confined spaces [9]. As devices shrink to the micro and nanoscale, they encounter more significant heat dissipation challenges per unit volume. With improvements in the packaging density and performance of microelectronic devices, the power of integrated circuit chips has significantly increased. This development also presents challenges for thermal management in electronic devices [10]. Therefore, research on the boiling heat transfer phenomenon at small and confined scales with efficient cooling systems for applications is crucial, including studying pool boiling phenomena in shallow pools, microchannels, or other confined areas. [11]. For pool boiling at confined scales, the scale of bubbles is usually smaller and influenced by the geometric shape of the channel and surface properties. Furthermore, small spaces also promote changes in liquid wettability, resulting in new characteristics of thermal performance parameters such as heat flux density and CHF [12]. A deep understanding of pool boiling phenomena in small and confined spaces is not only important for optimizing existing technologies but also lays the foundation for the development of future novel thermal management materials and devices.

Numerical simulations in computational fluid dynamics (CFD) are highly feasible, offering controlled environments for precise analysis. They enable detailed insights into complex phenomena and, are more cost and time-efficient compared to experiments in some cases. In the CFD study of pool boiling, detailed observations can be made on the formation and evaporation processes of microlayers, as well as the behavior of three-phase contact lines (TCLs). These microscale phenomena known as part of bubble dynamics have a significant impact on macroscopic HTC [13]. Classic numerical models commonly include macroscopic algorithms such as Reynolds-Averaged Navier–Stokes (RANS), Large eddy simulation (LES), and Direct Numerical Simulation (DNS). Compared to other simulation models, Lattice Boltzmann Method (LBM) extends feasible research from the continuum to the statistic description on a mesoscale. LBM models the fluid consisting of particles. These particles undergo consecutive propagation (streaming) and collision processes over a discrete lattice. LBM offers significant advantages over traditional CFD methods, particularly in integrating thermodynamics and simplifying computations for incompressible flows. It bypasses the resource-intensive process of solving the Laplace equation to satisfy the continuity equation, making it more efficient, especially for unsteady problems [14]. In this way, LBM possesses characteristics such as high parallelism, adaptability, and developability by solving the particle distribution function and recovering macroscopic parameters [15].

Many studies focused on using LBM to investigate the effects of surface characteristics and structures on bubble dynamics and macroscopic heat transfer performance. Ahmad et al. [16] used a 2D phase-change LBM to analyze pool boiling on a superheated plate with hybrid surfaces featuring cavities and pillars. The presence of pillars increased the surface area and altered bubble dynamics. Key findings included that enhanced bubble departure diameter and heat flux were achieved by adjusting pillar spacing and width, and surfaces with mixed wettability offered higher heat flux compared to uniformly hydrophilic surfaces with minimal increase in bubble size. Yu et al. [17] explored bubble dynamics and dry spot formation in boiling on a structured surface using a 3D LBM. The results showed that boiling efficiency depended on bubble departure and dry area, affected by secondary pillar size. Larger pillars increased bubble frequency but also dry spots. Reducing contact on pillar walls helped minimize dry spots. Optimal results occurred with

high bubble frequency and minimal dry spots. Wang et al. [18] used a 3D LBM to analyze boiling on large surfaces with different structural designs. They found that increasing the spacing between pillars helped bubble formation and heat transfer, while very tall pillars had the opposite effect. Wider pillars generally improved performance, but excessive width led to bubbles merging. Pillar–pillar structures performed better in heat transfer than pillar-pit ones due to better rewetting and thinner gas films. Oh et al. [19] introduced a new interfacial pseudopotential-based wetting method to overcome the limitations in capturing the unique features of non-isothermal boiling. The method incorporated mean fluid density at the solid–fluid interface and integrated molecular dynamics (MD) characteristics into the LBM framework. Results from multiphase simulations demonstrated the superior performance of the interfacial method in handling boiling phenomena, especially in capturing thermal gradients and interfacial dynamics. These studies focused on using LBM to investigate the influence of surface features, especially structured surfaces, on macroscopic heat transfer performance, providing insights into the relationship between surface structure or flow field and thermal field with macroscopic heat transfer coefficients. However, they lacked statistically significant contact line dynamics data to connect bubble dynamics parameters with macroscopic heat transfer behavior.

Recently, with the significant improvement in computing power, machine learning (ML) technologies have shown a remarkable developments in multiple fields, including data science, healthcare, finance, autonomous systems, as well as many other areas in science and engineering. Regarding boiling heat transfer, ML methods are being increasingly employed to understand and predict complex thermal phenomena. Many works have already been published that provide comprehensive reviews on the application of ML methods to the field of thermal fluid and boiling heat transfer [20–23]. The roles of ML methods in the field of thermal fluid dynamics and boiling heat transfer primarily involve: combining ML with CFD to optimize models and improve simulation efficiency; establishing data-driven predictive models for feature extraction and/or selection; as well as to assist in data analysis and image processing. Some works investigate the synergistic integration of ML methods with CFD frameworks, emphasizing the application of ML techniques within established CFD frameworks and the incorporation of CFD principles into ML predictive models. This integration aims to leverage the strengths of ML, thereby enhancing the computational efficiency and predictive accuracy of CFD. Chang et al. [24] explored five different methods for developing data-driven thermal fluid simulation models with ML methods: physics-separated; physics-evaluated; physics-integrated; physics-recovered; and physics-discovered. The study particularly emphasized the physics-integrated ML framework, which incorporated partial differential equations into the training of models, which highlighted the potential of this approach for leveraging big data. Panchigar et al. [25] examined various scenarios for the application of CFD and identified essential components necessary for effective computations in CFD. They highlighted that due to the unpredictable nature of fluid behavior, a single algorithm might not have been universally applicable; thus, combining multiple algorithms was often required for successful simulations. The authors also noted that the immaturity of ML techniques limited their ability to achieve high accuracy without significant preprocessing through numerical methods. Many studies have been conducted to enhance the computational efficiency of CFD simulations through ML approaches [26–28]. These works demonstrated how ML techniques can optimize various aspects of numerical modeling, streamline calculations, and ultimately reduce the time and resources needed for accurate simulations. This integration of ML techniques holds significant potential for enhancing the efficiency and accuracy of simulations in various fields, particularly in boiling heat transfer.

Existing studies that apply ML methods for direct prediction often rely on extensive databases to predict specific thermodynamic parameters, such as CHF and HTC. These databases compile an abundance of experimental data and numerical simulations, providing a robust foundation for training ML models. By analyzing the data, researchers can identify key factors influencing these parameters and develop more accurate and efficient predictive models, which can significantly enhance the understanding and performance of thermal systems. Li et al. [29] reviewed various methods for studying CHF in nucleate boiling systems and noted the potential of ML methods to significantly improve prediction accuracy compared to traditional empirical correlations. ML techniques showed lower root mean square error, indicating enhanced performance, but the application of these methods is still in development. The authors emphasized the need for comprehensive studies to optimize ML models, and the development of explainable AI that effectively balances accuracy and interpretability. An effective approach for predicting HTC is demonstrated in the work of Kalpana [30], which introduced a hybrid ML model that is effective in predicting HTC for complex geometries by integrating convolutional neural networks (CNNs) for geometric feature extraction with physics-informed neural networks (PINNs) to enforce physical constraints. The model demonstrated exceptional generalization capabilities and computational efficiency. There are other studies that focus on specific scenarios or dynamic parameters within the field of boiling heat transfer [7,31–33]. The application of data-driven ML methods in boiling heat transfer typically targets specific scenarios or dynamic parameters by constructing comprehensive databases that contain diverse experimental and simulation data. These ML methods are employed as alternatives to traditional empirical correlations or complex CFD models, enabling the identification of relationships between input variables and output parameters.

An increasingly common application of ML methods is in the field of computer vision, especially in areas including object detection and multi-object tracking (MOT), which has led to breakthroughs in various applications such as autonomous driving, surveillance systems, medical image analysis, and augmented reality. Especially in pool boiling, which involves the generation of numerous bubbles, ML methods have demonstrated significant potential in image analysis and pattern recognition. These methods can effectively identify and track the dynamic behavior of the bubbles, thereby providing quantitative data on bubble formation, growth, and detachment processes. By processing image data obtained from simulation, high-frequency video and thermal imaging, ML algorithms can extract key features and predict variations in heat flux and heat transfer coefficients. Zhou et al. [34] introduced a novel AI-based method for detecting and tracking condensing bubbles in subcooled flow boiling using object detection techniques. The model effectively captured key thermal–hydraulic parameters such as aspect ratio and bubble lifetime, achieving about 90% detection accuracy, while demonstrating good agreement with empirical correlations. In Torisaki's work [35], an AI-based object detection approach using CNNs was developed to extract real-time bubbly flow features from two-phase flow images. This method allowed for the instantaneous calculation of parameters such as void fraction, bubble number density, and average bubble sizes. The proposed technique proved effective at calculating these parameters in low void fraction regions. In the study by Wang et al. [36], a novel method was introduced for recognizing and tracking bubbles in a plate heat exchanger using a combination of CNNs and an improved three-frame difference method. The approach successfully measured bubble positions, velocities, and dimensionless parameters, achieving high performance metrics, including an average precision rate of over 94%, a recall rate of 87%, and an F1 score of 0.91, even under dense bubble conditions. ML methods offer new insights into the heat transfer mechanisms in pool boiling and aid in optimizing relevant thermal management systems and device designs.

In this paper, a hybrid approach combining LBM and ML methods is adopted to study the macroscopic heat transfer characteristics of surface properties and the microscopic contact line dynamics behavior in the pool boiling process. The boiling phenomena is confined to a

small space. LBM is used to investigate the effect of three surface properties (including surface superheat, wettability, and roughness) on the bubble nucleation process and heat transfer behavior. ML and computer vision techniques are employed to extract global contact line dynamics characteristics from LBM results. This analysis focuses on the influence of surface properties on contact line dynamics parameters and examines how these microscopic dynamics impact macroscopic heat transfer behavior. This study explores how surface parameters impact bubble nucleation and heat transfer behavior. ML methods are utilized to analyze the microscale contact line dynamic parameters of the entire bubble cluster statistically. The goal is to establish a relationship between surface parameters' effects on microscopic contact line dynamics and macroscopic heat transfer behaviors. Given the challenges of obtaining such large volumes of bubble dynamics data using conventional methods, the ML techniques are particularly valuable.

2. Hybrid thermal LBM

By discretizing the Boltzmann equation and considering the forcing term F_i, the Lattice Boltzmann equation can be deduced as:

$$f_i\left(\boldsymbol{x}+\boldsymbol{e}_i\Delta t, t+\Delta t\right) = f_i(\boldsymbol{x},t) + \Omega_i(\boldsymbol{x},t) + F_i(\boldsymbol{x},t) \tag{1}$$

where f_i denotes the particle distribution function along the ith lattice direction, $\boldsymbol{x}$ denotes the spatial position, $\boldsymbol{e}_i$ denotes the discrete velocity, t denotes the time, and F_i denotes the force. Ω_i is the collision operator and makes f_i approach the equilibrium particle distribution function f_i^{eq}. For an isothermal simplification, f_i^{eq} is given as:

$$f_i^{\text{eq}} = w_i\rho\left(1+\frac{\boldsymbol{u}\cdot\boldsymbol{e}_i}{c_s^2}+\frac{\left(\boldsymbol{u}\cdot\boldsymbol{e}_i\right)^2}{2c_s^4}-\frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right) \tag{2}$$

where w_i denotes the weight corresponding to the lattice and $\boldsymbol{e}_i$, ρ denotes the density, $\boldsymbol{u}$ denotes the velocity, and c_s denotes the lattice sound speed. A simple collision operator, Bhatnagar, Gross and Krook (BGK) collision operator $\Omega(f) = -\frac{1}{\tau}(f-f^{eq})$ is used in this work, where τ represents the relaxation time. For a simple two-dimensional LBM problem, the velocity set *D2Q9* is the classic paradigm for hydrodynamics [37].

The forcing term F_i is defined as [38]:

$$F_i = w_i\left[A_0+\frac{\boldsymbol{B}\cdot\boldsymbol{e}_i}{c_s^2}+\frac{\boldsymbol{C}:\left(\boldsymbol{e}_i\boldsymbol{e}_i-c_s^2\boldsymbol{I}\right)}{2c_s^4}\right] \tag{3}$$

Here, $A_0 = 0$, $\boldsymbol{B}$, and $\boldsymbol{C}$ are parameters used for different force schemes. Typically, $A_0 = 0$ to conserve mass, while $\boldsymbol{B} = B_e\boldsymbol{F}$, and $\boldsymbol{C} = C_e(\boldsymbol{vF}+\boldsymbol{Fv})$ to recover the Navier–Stokes equations. Among the many forcing methods, Guo's scheme [38] has gained significant popularity due to its systematic approach and thorough analysis of the compulsory scheme, successfully avoiding undesirable artificial outcomes in momentum caused by discretization.

In the field of multiphase flow using LBM, the Shan–Chen method [39,40], also known as the pseudopotential method, is primarily focused. This method aims to incorporate non-local interactions for integrating non-ideal equations of state. The implementation of this method relies on introducing a local attraction forcing term that depends on the surrounding non-local density. This forcing term helps maintain uniform density in the surroundings, leading to the emergence of different phases. The pseudopotential method is popular due to its simplicity and mesoscopic origin, and has been widely used in various simulations such as droplet impact on surfaces [41] and splashing [42]. A classic implementation is shown as follows:

$$\boldsymbol{F}(\boldsymbol{x}) = -G\psi(\boldsymbol{x})\sum_i w_i\psi\left(\boldsymbol{x}+\boldsymbol{e}_i\right)\boldsymbol{e}_i \tag{4}$$

where ψ refers to the pseudopotential function, G denotes the interaction strength coefficient, and w_i refers to the weight of the velocity direction. The pseudopotential function ψ is defined as followed, related to the pressure p which can be defined by the state equation:

$$\psi(\rho) = \sqrt{\frac{2\left(p-\rho c_s^2\right)}{c_s^2 G}} \tag{5}$$

The Peng–Robinson (PR) equation of state (EoS) is used for calculating the pressure. The PR EoS is well-suited for describing multiphase water flow [43,44]:

$$p = \frac{\rho RT}{1-b\rho} - \frac{a\rho^2\alpha\left(\omega_{\text{EoS}}\right)}{1+2b\rho-b^2\rho^2} \tag{6}$$

$$\alpha\left(\omega_{\text{EoS}}\right) = \left[1+\left(0.37464+1.54226\omega_{\text{EoS}}-0.26992\omega_{\text{EoS}}^2\right)\left(1-\sqrt{\frac{T}{T_c}}\right)\right]^2 \tag{7}$$

where $\omega_{\text{EoS}} = 0.344$ and a commonly used set of parameters is $a = \frac{2}{49}$, $b = \frac{2}{21}$, and $R = 1$ [44] and is used in our work.

This work uses the double distribution function (DDF) method in thermal LBM to illustrate the evolution of temperature fields. The DDF method [37,45,46] couples convection and diffusion of heat and momentum, enabling energy introduction into LBM. In this work it is based on a semi-hybrid thermal method [47] to introduce the phase transition in the liquid lattice and the latent heat in the thermal lattice. The phase change term is based on literatures [48,49], while the semi-hybrid approach is the same as Kamali's work [50]:

$$g_i\left(\boldsymbol{x}+\boldsymbol{e}_i\Delta t, t+\Delta t\right) - g_i(\boldsymbol{x},t) = -\frac{1}{\tau_g}\left(g_i-g_i^{\text{eq}}\right)+C_i+\left(G_i+\frac{1}{2}\frac{\partial G_i}{\partial t}\right) \tag{8}$$

$$C_i = \left(1-\frac{1}{2\tau_g}\right)\frac{w_i\boldsymbol{e}_i\cdot\frac{\partial(T\boldsymbol{u})}{\partial t}}{c_s^2} \tag{9}$$

where g_i denotes the temperature distribution function for the secondary lattice. G_i is the source term and is shown as follows:

$$G_i = w_i q$$
$$q = \frac{1}{\rho c_v}\nabla\cdot(\lambda\nabla T)-\nabla\cdot(D\nabla T)+T\left[1-\frac{1}{\rho c_v}\left(\frac{\partial p}{\partial T}\right)_\rho\right]\nabla\cdot u \tag{10}$$

In this work, numerical simulation is carried out on the open-source LBM simulation platform OpenLB [51]. The simulation uses a pseudopotential thermal multiphase extension developed by Delft University of Technology in the Netherlands [52]. The DDF method is introduced to couple momentum and heat transport in LBM based on a semi-hybrid thermal method. The developer of this model evaluated its performance in various aspects including the Maxwell construction, thermodynamic consistency, Young–Laplace law, wall wettability, and D^2 law in the previous work [52], proving that the model has the ability to simulate multiphase flow with phase transition using LBM.

3. LBM simulation of pool boiling on a flat surface

3.1. Impact of roughness and surface density on surface wettability

LBM describes the heated surface through a bounce-back (BB) scheme. For no-slip flow, incoming particles are bounced back in the same direction, while for slip flow, incoming particles are reflected like light. The BB scheme assumes that the surface is located halfway between nodes and operates on the particle distribution function by introducing ghost nodes inside the surface. In order to describe surface wettability, an additional force between the surface and the fluid near the surface is added on top of the BB scheme to simulate interactions between the surface and fluid. This study models this force as follows with a definition similar to that of Eq. (4) [53]:

$$\boldsymbol{F}_s(\boldsymbol{x}) = -G\psi(\boldsymbol{x})\sum_i w_i\psi\left(\boldsymbol{x}+\boldsymbol{e}_i\Delta t, t\right)\boldsymbol{e}_i \tag{11}$$

Table 1
The impact of spatial resolution N_x on calculation results.

Spatial resolution N_x	Contact angle θ (°)		
	ξ = 2.0	ξ = 5.0	ξ = 7.0
150	105.92	49.55	16.36
75	107.94	50.40	18.18
50	112.62	51.12	Not converging

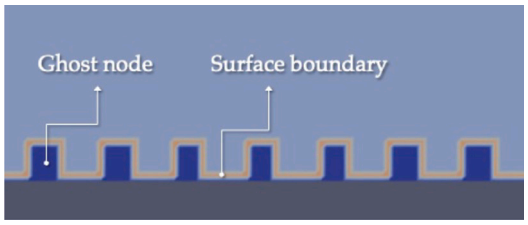

Fig. 1. Structured surface representing roughness.

where $G = -1$ and $\psi(\boldsymbol{x} + \boldsymbol{e}_i \Delta t) = \psi(\xi)$ is the lattice version of the mean-field potential. This surface potential is a free parameter in the model and should not be linked to the actual density of the solid phase. The parameter ξ in $\psi(\xi)$ is used to tune different surface wettability as a *fictitious surface density* [40,54].

Roughness is also studied as an additional factor. A serrated structure is added to simulate a structured surface with microscale roughness, as illustrated in Fig. 1. All the outer outlines of the surface structure are approximately square. The arithmetic average roughness, Ra, is used to evaluate surface roughness. The calculation of the droplet contact angle θ is presented as follows:

$$\theta = \arctan\left(\frac{l_\mathrm{d}}{2\left|\frac{4h_\mathrm{d}^2 + l_\mathrm{d}^2}{8h_\mathrm{d}} - h_\mathrm{d}\right|}\right) \tag{12}$$

where l_d and h_d represent the contact length with the surface and the height of the droplet, respectively.

The calculation domain has dimensions of 1000 μm in length along the x-axis and 100 μm in height along the y-axis. The simulations are initialized with a smooth sphere indicator representing the droplet, positioned in the middle of the x length, just above the surface. The sphere has a radius equal to one sixth of the domain height. The boundaries in the x direction are periodic, and the surface is set as a no-slip boundary. We first verify the possible impact of spatial resolution N_x on the computation results. Simulations are repeated using three spatial resolutions $N_x = 50, 75, 150$, and the contact angles calculated from Eq. (12) at three surface densities are obtained. The results, presented in Table 1, indicate that both $N_x = 150$ and $N_x = 75$ yield very close contact angles, while further reducing N_x to 50 fails to achieve convergence within a limited time. Considering both the time cost and the required precision of the results, the calculations in this section are based on $N_x = 150$, while those in the next section are based on $N_x = 75$.

The relationship between wettability and surface density under different roughness conditions is depicted in Fig. 2. The results indicate that across all roughness levels, the contact angle decreases with an increase in surface density. The outcomes for samples with roughness are compared with theoretical data available in the literature [53], demonstrating that surface wettability can be modified by altering ξ. As ξ increases from 0.5 to 7.5, the surface transitioned from hydrophobic to hydrophilic.

The introduced roughness also adds to the complexity of the relationship between the contact angle and ξ; specifically, it increases the contact angle at smaller or larger surface densities, while reducing it at moderate surface densities. Also, changes in surface density lead to

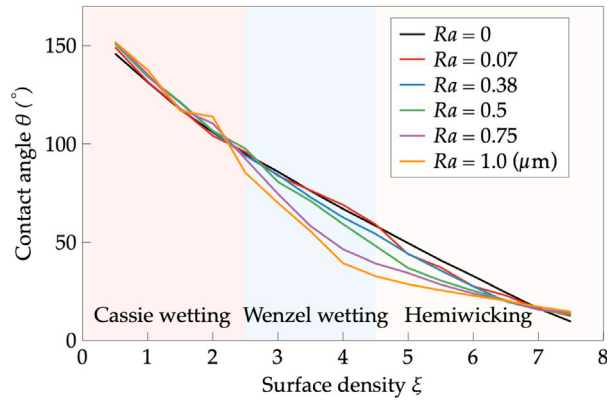

Fig. 2. Relationship between contact angle θ and surface density ξ under different surface roughness Ra, with background colors indicating wetting modes.

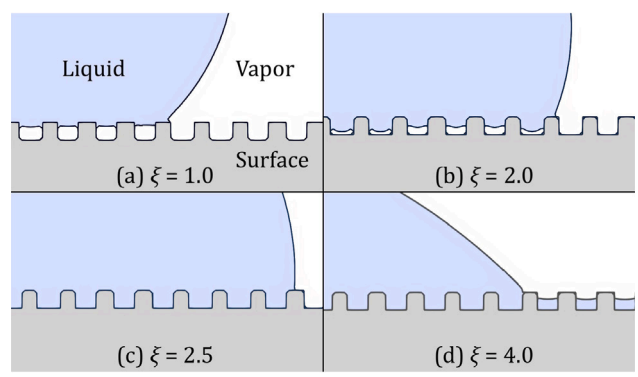

Fig. 3. Wetting modes under different surface densities ξ at Ra = 0.8 μm.

variations in the wetting properties exhibited by droplets on a surface. The simulated results under typical wetting conditions are shown in Fig. 3 after processing to obtain clear outlines. Under almost all roughness conditions, at surface densities below 2.0, vapor gaps exist between the droplet and surface, resulting in incomplete wetting analogous to the *Cassie–Baxter state*. As the surface density increases above 2.5, these gaps begin to fill, transitioning towards the *Wenzel state*. When surface density exceeds 4.5, the penetration front extends beyond the droplet boundary, a phenomenon known as *hemiwicking*. Surface density therefore influences not only the contact angle but also the wetting modes. Employing different surface densities in LBM simulations facilitates the mimicry of varying the contact angle and wetting modes.

3.2. Impact of surface characteristics on bubble nucleation and heat transfer

A two-dimensional LBM simulation is performed to study bubble nucleation on a heated surface and to analyze the impact of surface characteristics. The spatial resolution and temporal resolution of the model are 75 and 7, respectively. The characteristic length L_ch is 1.0×10^{-4} m and the characteristic speed u_max is 20 m/s. The fluid is set as water. The simulation domain is a rectangle with dimensions 3.2×10^{-3}

m by 6.0×10^{-4} m along the x and y-axes, respectively. A bounce-back (BB) scheme is used for the bottom heated surface and the upper surface, and these two surfaces stay constant temperature throughout the simulation. The left and right boundaries are periodic boundary conditions. In LBM, for no-slip flow, incoming particles are bounced back in the same direction; while for slip flow, incoming particles are reflected like light. The BB scheme assumes that the surface is located halfway between nodes and operates on the particle distribution function by introducing ghost nodes inside the surface. At the beginning of the simulation, the entire computational domain undergoes an initialization process of 5000 steps, and the lower 40% of the area is initialized as liquid, while the rest is vapor. The entire computational domain is initialized according to the initial temperature $T_{in} = 0.7 * T_c$ during the initialization, where T_c is the critical temperature of water. After the initialization is completed, the surface temperature T_w is reset according to the conditions and an additional random temperature fluctuation is added to induce bubble nucleation. The random temperature fluctuation is defined by the random function of the C standard library and its range is set within $2.5 * T_c$. Weinmiller [52] specifically discussed the issue of activating bubble nucleation on the surface, while we follow other previous works [55–57] by using this surface heat spot to induce bubble nucleation. The thermal physical parameters are selected based on liquid water at 100 °C and vapor at 115 °C. The node interval is 1.3×10^{-6} m, the timestep Δt is 9.52×10^{-9} s, and data are collected every 20,000 Δt after the initialization. The entire simulation process lasts 0.005 s (simulation time).

Three surface-related characteristics are investigated for their effects on bubble nucleation and heat transfer behavior. The first is the surface superheat, represented by ΔT^* as followed:

$$\Delta T^* = \frac{T_w - T_{in}}{T_c} \tag{13}$$

where T_w denotes the surface temperature, $T_{in} = 0.7 * T_c$ denotes the initial fluid temperature, and T_c is the critical temperature of water. The second characteristic is surface density ξ, which affects both surface wettability and wetting modes as previously mentioned. In the simulation of this section, ξ ranges from 2.0 to 7.0, corresponding to contact angles for ideal smooth surface of about 106° to 16° in Fig. 20. Smaller ξ values (more hydrophobic) greatly limits bubble nucleation, while larger ξ values (more hydrophilic) causes computational instability. Consequently, these values are excluded from the simulation. The third characteristic is surface roughness, which is introduced in accordance with Fig. 1 and represented by Ra. To describe bubble nucleation quantitatively, a reference line $y = 2.0 \times 10^{-5}$ m is arbitrarily selected. The density distribution along this line is analyzed, and 30% of the maximum density on this line is taken as the boundary between the vapor and liquid phases, with continuous vapor phases treated as a single bubble. The bubble number N_b and vapor phase ratio, that is, void fraction ϕ along this line are analyzed to provide a simple visual representation of bubble nucleation:

$$\phi = \frac{\text{Reference line length occupied by the vapor phase}}{\text{Reference line length}} \tag{14}$$

The influence of surface characteristics on heat transfer is further studied. The spatially averaged heat flux q and HTC h is calculated using the fluid temperature on the reference line and surface temperature:

$$q(t) = \int_0^L \frac{q_w(x)\mathrm{d}x}{L} = \int_0^L \frac{-\lambda \left.\frac{\partial T_f}{\partial y}\right|_{y\to 0} \mathrm{d}x}{L} \tag{15}$$

$$h(t) = \frac{q(t)}{T_w - T_s} \tag{16}$$

where T_f, T_w, T_s, L, λ represent fluid temperature, surface temperature, saturation temperature, total horizontal length (the length of the domain in the x direction, 3.2×10^{-3} m), and thermal conductivity, respectively. We use h to represent heat transfer changes in the fluid near the surface during simulation and time-averaged q to analyze how surface parameters impact heat transfer.

3.2.1. Effects on bubble nucleation and void fraction

A typical pool boiling simulation result is shown in Fig. 4 under the condition that $\Delta T^* = 0.20$ and $\xi = 5.0$. The density is in lattice unit. Roughness is not considered. After the initialization, bubbles begin to form on the surface. The bubbles continue to grow and start to coalesce. Due to the small size of the vertical dimension, which is much smaller than the typical bubble departure diameter, bubbles do not depart but rather break up upon contact with the upper boundary. The nearby bubbles move towards the gap, and vacant areas continue to generate bubbles. As the bubble nucleation is induced by local hot spots, the surface temperature distribution is measured after the initialization step. The result shows that the surface superheat after initialization of this case can correspond to a superheat of 3.0 K in actual situations.

The bubble number N_b and void fraction ϕ vary with ΔT^*, as shown in Figs. 5 and 6. Blue histograms show the evolution of N_b and ϕ over time. $N_{b,max}$ and $\overline{\phi}$ refer to the maximum value of N_b and the time-averaged value of ϕ, respectively. As the maximum value, $N_{b,max}$ is used to evaluate the number of bubble nucleation sites on the surface, and the time-averaged values are used to evaluate the whole process. For the typical case with $\Delta T^* = 0.225$ in Fig. 5, the change in N_b intuitively reflects dynamic behaviors of bubble nucleation. At the beginning of the simulation, the number of bubbles increases rapidly and quickly reaches its peak $N_{b,max}$ around $40,000\Delta t \sim 60,000\Delta t$. As bubbles grow, adjacent ones coalesce, resulting in a decrease in N_b. When a large bubble grow beyond the vertical boundary, it ruptures and quickly gets filled by nearby liquid, where bubble nucleation should occur again. In most cases, the simulation time ensures that multiple cycles of dynamic bubble nucleation, coalescence, rupture, and re-nucleation are covered before reaching a steady state.

The results indicate that significant bubble nucleation occurred only at $\Delta T^* > 0.20$. When $\Delta T^* = 0.15$, although nucleated bubbles gradually grow (ϕ continued to rise for some time), there are always limited nucleation sites. When the surface superheat exceeded $\Delta T^* \geq 0.25$, $N_{b,max}$ further increased significantly, reflecting the enhancement of the surface bubble nucleation ability after initialization due to high surface temperature. Increasing the surface superheat beyond would not further increase $N_{b,max}$, but it would cause the time-averaged void fraction $\overline{\phi}$ to rise. Increasing ΔT^* from 0.2 to 0.25 and then to 0.30 respectively results in a 30% increase in $\overline{\phi}$. Further increasing ΔT^* to 0.35 only lead to a 15% increase in $\overline{\phi}$. Taking into account that the further significant increase in $\overline{\phi}$ indicates a larger space occupied by bubbles, this trend shows the increase in the moving speed of bubble TCLs, without observing a significant increase in N_b and $N_{b,max}$ with increasing ΔT^*. Although using N_b can describe the bubble behaviors more intuitively, its small quantity and the randomness of nucleation process may introduce instabilities. Based on the approximate trend shown by ϕ and N_b, as well as its stronger robustness, we use ϕ and specific bubble behaviors to describe the bubble nucleation process near the surface in subsequent studies. The effect of surface roughness Ra on the bubble nucleation process and void fraction is further studied. The implementation of surface roughness is similar to Fig. 1, with a serrated structure uniformly distributed on the surface to simulate a structured surface with roughness. Ra increases from 0 (ideal smooth surface) to 10.0 μm. For evaluating bubble nucleation and heat transfer, the same reference line $y = 2.0 \times 10^{-5}$ m is used, but the baseline of $y = 0$ is assumed to be the upper surface of the serrated structure. The relationship between ϕ with Ra is shown in Fig. 7. The results show that $\overline{\phi}$ exhibits a trend of first decreasing and then increasing with rising Ra. The minimum value is found to occur around $Ra = 0.8$ μm, while the maximum value is around $Ra = 7.0$ μm. Compared to a smooth surface, smaller Ra reduces $\overline{\phi}$,weakening the effect of keeping bubbles on the surface. This may be due to our choice of reference line, which allows for more vapor between surface structures. Another significant feature is the strong fluctuation in the evolution of ϕ, indicating intense bubble coalescence and re-nucleation processes. The introduction of Ra brings more nucleation sites and instability in TCLs motion. However,

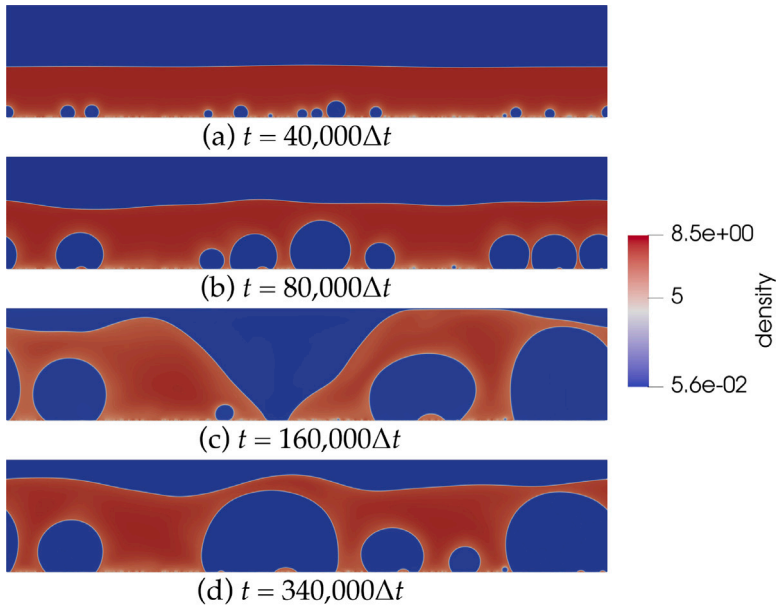

Fig. 4. Lattice density at different Δt under condition of $\Delta T^* = 0.20$, $\xi = 5.0$, $Ra = 0$.

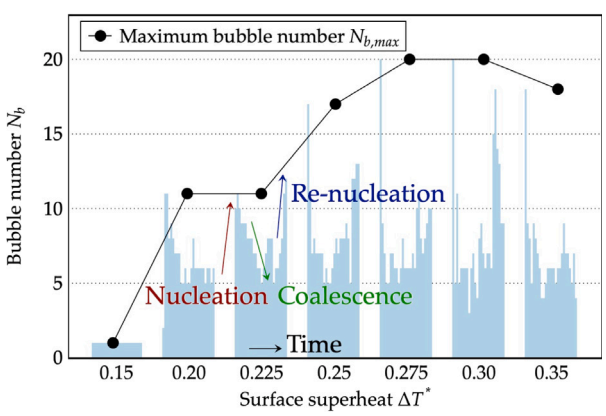

Fig. 5. Evolution of N_b (blue histograms, moving from left to right representing the simulation from start to end, same as below) and its maximum value $N_{b,max}$ (line with dots) with ΔT^* under the condition of $\xi = 5.0$ and $Ra = 0$.

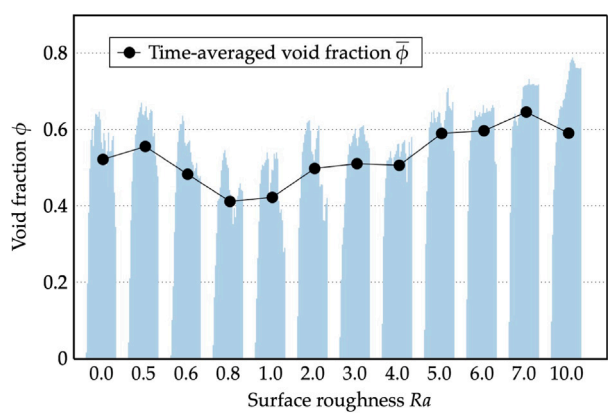

Fig. 7. Evolution of ϕ (blue histograms) and its time-averaged value $\overline{\phi}$ (line with dots) with Ra under the condition of $\Delta T^* = 0.20$ and $\xi = 5.0$.

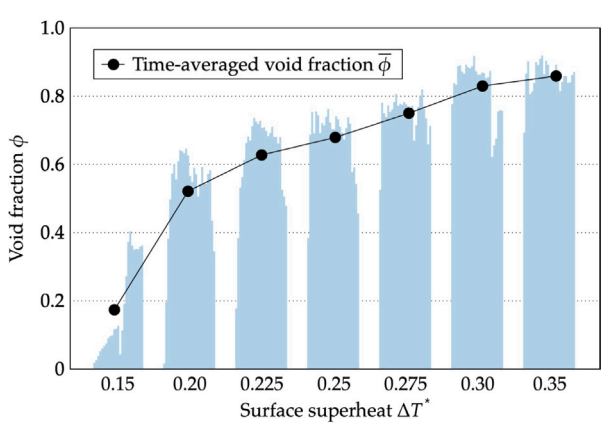

Fig. 6. Evolution of ϕ (blue histograms) and its time-averaged value $\overline{\phi}$ (line with dots) with ΔT^* under the condition of $\xi = 5.0$ and $Ra = 0$.

when Ra exceeds a certain threshold it has the opposite effect. This may be caused by the *pinned effect* of TCL motion implemented on a larger Ra surface. Fig. 8 illustrates a common occurrence where the right boundary of a bubble is pinned due to the influence of structured surface. No re-nucleation occurs in this case with $Ra \geq 5.0$ μm according to our data of bubble number. This pinned effect of the structural surface on bubble growth prolongs the survival time of large bubbles on the surface, increasing the $\overline{\phi}$ while reducing the possibility of bubble coalescence and future re-nucleation. Additional evidence is seen in the consistent trend of ϕ at high Ra values. For cases where $Ra \geq 5.0$, ϕ shows greater stability compared to around $Ra = 0.8$, indicating an improved stability in the bubble nucleation process.

The effect of surface density ξ on the bubble nucleation process and heat transfer under different Ra is also studied, as shown in Fig. 9. As mentioned above, lower ξ corresponds to a more hydrophobic surface with less complete wetting, while a higher one corresponds to a more hydrophilic surface with more complete wetting. Accordingly, the results of ξ well demonstrate the variation of surface wettability; as ξ increases, $\overline{\phi}$ decreases, indicating a reduction in the surface's ability to bind with the vapor phase. For the smooth surface, $\overline{\phi}$ and ξ show an approximate linear relationship. For every decrease of 1.0 in ξ, $\overline{\phi}$ decreases by about 5%. For $\overline{\phi}$, it is noted that for different values of Ra, $\overline{\phi}$ shows a consistent correlation with ξ. In almost all cases of the same ξ, introducing Ra initially causes $\overline{\phi}$ to decrease and then

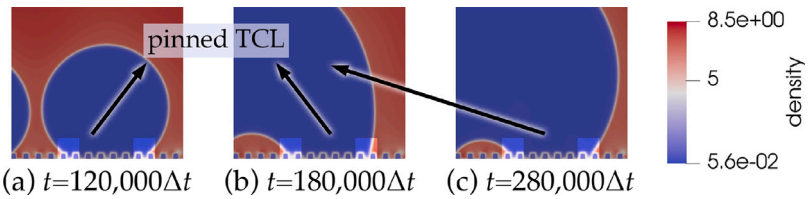

Fig. 8. Typical phenomenon of bubbles being pinned under the condition of $\Delta T^* = 0.20$, $\xi = 5.0$, $Ra = 5.0$ μm.

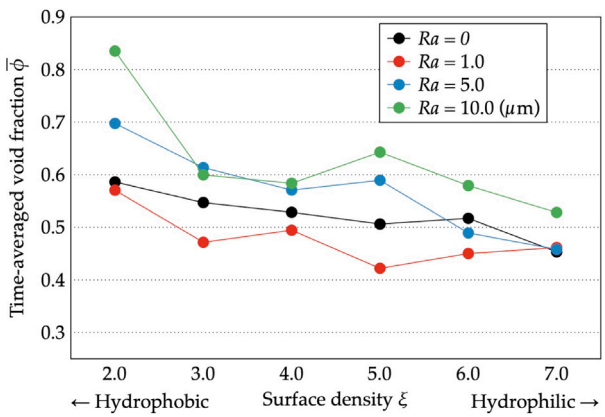

Fig. 9. Relationship between $\overline{\phi}$ with ξ under $\Delta T^* = 0.20$ and various Ra.

increase again. A surface with $Ra = 1.0$ causes a decrease in $\overline{\phi}$ on surfaces of different wettability compared to a smooth surface. From the perspective of the influence of Ra on surface wettability, small Ra enhances surface wetting, but further increasing Ra would reduce this effect after exceeding specific threshold value. The void fraction is an intermediate value used to analyze nucleation behavior; however, it plays a crucial role in enhancing our understanding of how surface parameters affect heat transfer. The effect of three surface parameters on the near-surface void fraction ϕ and its time- and spatially averaged value $\overline{\phi}$ are shown in this section, but their mechanisms are different. Combining our bubble number data, the influence of ΔT^* on $\overline{\phi}$ is mainly in terms of bubble nucleation ability. The higher surface nucleation ability enhanced by a higher ΔT^* significantly increases $\overline{\phi}$. For $\overline{\phi}$, ξ is more intuitively associated with the wettability of the surface to bind with vapor (or liquid phase) phase. Ra's influence on $\overline{\phi}$ is more complex. We choose to interpret this impact by combining the temporal evolution of ϕ with specific bubble behaviors.

3.2.2. Effects on heat transfer

The evolution of HTC h and heat flux q with different ΔT^* is shown in Fig. 10. As ΔT^* rose from 0.15 to 0.20, there is a step increase in q, similar to the increasing behavior of $\overline{\phi}$ in Fig. 6. After ΔT^* exceeds 0.20, the effect of increasing ΔT^* on q is weakened. The maximum value of h generally occurs right after the initialization phase ends. This is when the temperature difference between the surface and liquid is at its peak, and there is sufficient space, leading to rapid bubble nucleation and the coalescence of tiny bubbles. This causes a decrease in N_b as shown in Fig. 5 simultaneously, thus driving flow. The combined effects of the high temperature difference, as well as the flow caused by rapid coalescence, results in h reaching its peak before quickly decreasing. For high ΔT^* values ($\Delta T^* \geq 0.275$), the peak value of h occurs immediately after initialization because the temperature difference has a dominant effect. For lower ΔT^*, bubbles rapidly coalesce after initialization, causing the peaking of h to occur later before further decreasing.

Another point worth noting is that for the cases of $\Delta T^* = 0.225$, 0.25, 0.275, and 0.30, h shows a further increase near the end of the simulation, corresponding to a more obvious re-nucleation process, as observed from N_b in Fig. 5. This rapid increase in h, followed by a decrease in N_b, is observed. At this time, some bubbles reach the upper boundary and break up, leading to a process of bubble re-nucleation, as well as the sudden rise of h. Taking the typical case of $\Delta T^* = 0.30$ as an example, there is a rapid decrease in N_b after initialization, which remains relatively low throughout and is also lower compared to other cases. Morphologically speaking, as shown in Fig. 11, several large bubbles divide the entire boiling region into segments throughout the entire simulation. These bubbles develop individually until one reaches the upper boundary and break up, causing disturbances that lead to successive ruptures of other bubbles, after which bubble re-nucleation occurs. This mode is also evident under different ξ values, as depicted in this figure. Additionally, this behavior can be observed in the evolution trend of ϕ, such as the relatively stable trend of ϕ at the early stage for $\Delta T^* = 0.30$, shown in Fig. 6. In this mode of less bubble coalescence, heat transfer is significantly suppressed. This results in q being lower than the cases of $\Delta T^* = 0.275$ and $\Delta T^* = 0.35$.

The evolutions of h and q with different Ra are shown in Fig. 12. With increasing Ra, q first increases to a certain extent and then decreases continuously. The maximum value of q is around $Ra = 1.0$ μm, and when $Ra > 5.0$ μm, q becomes lower than the value under $Ra = 0$. Compared to $Ra = 0$, $Ra = 1.0$ increases q by about 10%, while $Ra = 10.0$ reduces q by about 25%. The data shows that q reaches its maximum value when $Ra = 1.0$ μm. Following the initial peak value, h rapidly decreases before gradually rising to reach a second peak. This corresponds to the process in which ϕ gradually increases to a peak and then decreases in Fig. 7. Compared to the set of cases with a smaller Ra, it takes longer to reach the second peak value of h, and the subsequent decrease in h also takes longer, resulting in higher q. This well reflects the positive impact on h from enhanced microconvection caused by bubble growth and coalescence due to introduced Ra. The enhancing effect of Ra on h has also been confirmed in many experimental studies [58,59]. However, this impact does not continue to increase as Ra increases. Due to the structured surfaces present, movement of TCLs may be pinned at the edges of structures as we illustrated in Fig. 8. In this mode, it becomes more difficult for coalescence and future re-nucleation processes with high Ra, also demonstrated by the more stable evolution of $\overline{\phi}$ at $Ra = 6.0$ and 7.0 in Fig. 7, thus leading to lower q. More evidence is found in the case of $Ra = 5.0$, which represents the turning point where the heat transfer ability transitions from enhanced to diminished compared to a smooth surface, as illustrated in Fig. 12. As shown in Fig. 8, the high Ra structural surface prevents small bubbles coalesce, creating a pinned effect on the motion and a relatively high $\overline{\phi}$. As the liquid is heated, h constantly decreases from its peak value, and the lack of bubble coalescence and re-nucleation in the second half processes results in a lower q value. Similar situations also occur in cases when $Ra \geq 6.0$ μm. Higher Ra may increase the likelihood of TCLs being pinned in motion, causing individual bubbles to grow independently and ultimately decreasing q.

The effect of surface density ξ on heat transfer under different Ra is also studied as shown in Fig. 13. For the same ξ, adding roughness causes q to first rise and then fall beyond a specific range. For the four values of Ra involved in Fig. 12, q when $Ra = 1.0$ μm is always higher than that when $Ra = 0$ for all ξ, while with $Ra = 10.0$ μm, q is consistently lower. This property exhibits similarities under different

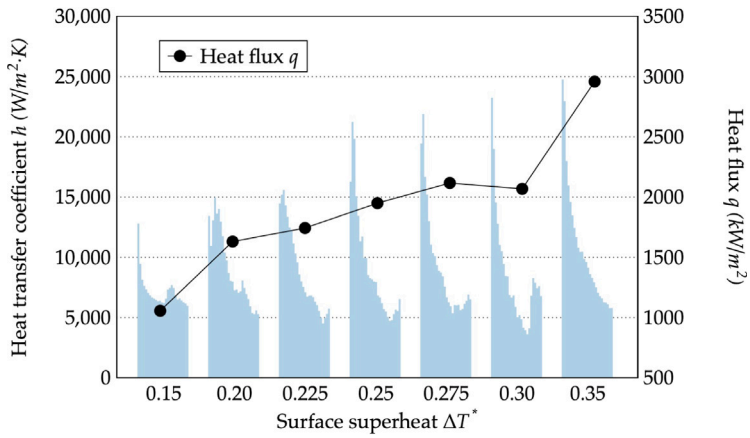

Fig. 10. Evolution of h (blue histograms) and q (line with dots) with ΔT^* under the condition of ξ = 5.0 and Ra = 0.

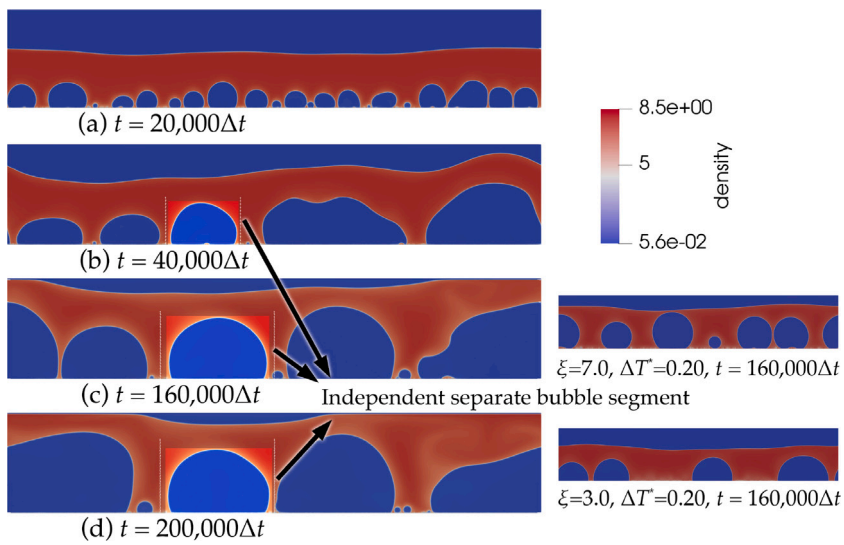

Fig. 11. Lattice density at different Δt under condition of ΔT^* = 0.30, ξ = 5.0, Ra = 0.

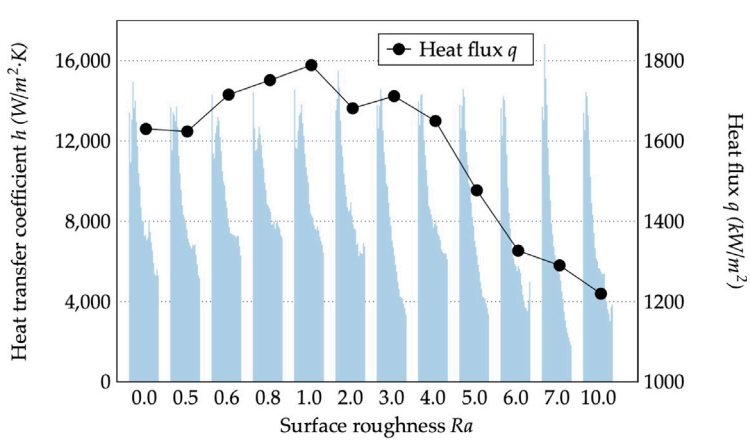

Fig. 12. Evolution of h (blue histograms) and heat flux q (line with dots) with Ra under the condition of ΔT^* = 0.20 and ξ = 5.0.

ξ. One interesting point is that if considering a structural surface with a heat transfer capacity similar to the smooth surface, the value of this Ra seems to be larger in the hydrophobic state and smaller in the hydrophilic state. After ξ exceeds 5.0, the increasing trend of q with ξ is weakened. Regardless of the value of Ra, increasing ξ from 5.0 to 7.0 does not result in a significant change in q compared to when ξ increases from 2.0 to 5.0. For all cases, the variation of q does not exceed 10% when $\xi \geq 5.0$. Combining the previous analysis of the effect of ξ on the wetting modes, namely that between 4.0 and 5.0 is the transition point from the Wenzel wetting state to hemiwicking, we conclude that in the hemiwicking wetting mode, changes in contact angle have smaller impact on q compared to others. Before entering

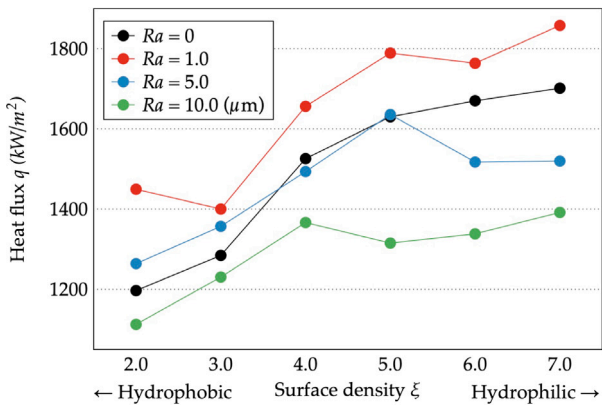

Fig. 13. Relationship between q with ξ under $\Delta T^* = 0.20$ and various Ra.

hemiwicking, q exhibits a more pronounced characteristic of increasing with larger ξ. In summary, for surfaces of the same wettability, the introduction of Ra first increases q (higher than a smooth surface) and then decreases it (lower than a smooth surface). For both rough and smooth surfaces, transitioning from hydrophobic to hydrophilic causes q to increase within a certain range and remain further after entering the hemiwicking wetting mode.

4. Data mining for contact line dynamics parameters combining machine learning and computer vision

In numerical simulation, conventional post-processing methods can effectively extract some core information for the obtained time-dependent vector fields or scalar fields. When the research focus shifts to more complex transient phenomena or disordered system characteristics, these methods often struggle to provide sufficient insight. Especially in our study of the pool boiling process on flat surfaces, there may be a strong link between microscopic contact line dynamics data and macroscopic thermal or flow data, and revealing this connection is crucial for a deeper understanding of boiling heat transfer. Here we adopt a data mining process that combines machine learning and computer vision technologies to handle data obtained from simulations. Specifically, it includes tasks such as bubble detection, bubble tracking, individual separation, and extraction of contact line dynamics data. One important feature of this image- or video-based data mining process is that it is completely decoupled from the simulation process. This means that with different simulation configurations or experimental results, one does not need to modify the post-processing process; instead, by adjusting the input data, broader identification and extraction of contact line dynamics data can be achieved.

Specifically, we first use the YOLO (You Only Look Once) [60] model for bubble recognition. This phase is based on simulation results obtained by LBM. Next, we apply MOT technology to capture the behavioral changes of bubbles across different time frames, allowing us to continuously monitor the motion and changes of individual bubbles between multiple frames. Then for important contact line dynamics parameters, we adopt a composite image processing algorithm, separating the tracking individuals and obtaining data. This algorithm can effectively determine the boundary position of TCLs and estimate the contact angle. A simple workflow of this data mining process is shown in Fig. 14. The *supplementary material* provides additional information on the algorithm principles and technical details of bubble recognition and tracking.

It must be emphasized that, traditional CFD post-processing methods, such as the threshold segmentation method based on density, are also crucial for studying contact line dynamics. These methods especially help in analyzing the growth and detachment processes of individual bubbles by providing precise bubble boundaries. However, our research focuses on studying how surface parameters influence the speed, acceleration, and contact angle characteristics of bubble clusters during nucleation and coalescence. Traditional post-processing methods still require a large amount of complex work when dealing with these variables of bubble clusters. It is also difficult to avoid introducing subjective judgment factors especially when handling bubble coalescence issues. This is where the advantage lies in using ML post-processing methods in this section. Furthermore, our method is based entirely on image data derived from CFD results; therefore it is completely separate from CFD methods which allows this type of post-processing method to be applicable to a wider range of scenarios.

4.1. Pre-processing

Data from Section 3 is used to create the training image set. The dataset includes simulation data with varying surface density ξ as this parameter has the most significant influence on bubble morphology, thereby improving the prediction model's effectiveness. The open-source software, LabelImg [61] is utilized to establish associated labels from the images of LBM results for input into YOLOv8. This image labeling tool can generate annotations in multiple formats to suit different needs. Approximately 5000 bubbles are manually labeled using this method. Due to the relatively simple and single scene of the image dataset generated by simulation, we do not apply data augmentation in the subsequent process. Instead, a certain degree of extension is made for the left and right boundaries to capture bubbles located on the periodic boundaries. Subsequent validation data indicate that for bubble tracking in such simulation scenarios, the YOLO model generated already has a fairly high accuracy.

During the training process of the YOLO model, the setting of hyperparameters significantly impacts the predictive performance of the generated model. In order to achieve optimal model performance, we first study the arrangement of several important sets of hyperparameters on model prediction performance. The loss value is selected to evaluate the model's quality by minimizing losses in the YOLOv8 process, ultimately choosing the best model as the optimal detection model. Providing multiple pluggable state-of-the-art MOT trackers, BoxMOT offers a simple and efficient way to compare the performance of different trackers in specific MOT tasks. We compare the tracking capabilities of five MOT trackers and find that the ImprAssoc tracker [62] consistently outperforms the others in our application. As a result, we incorporate it into our post-processing workflow. In addition, we developed an algorithm to batch process bubble data to obtain speed, acceleration, and contact angle characteristics relevant to TCL dynamics. Considering the coherence of the paper, the related technical information is not included. Please refer to our supplementary material for more technical information.

4.2. Impact of surface characteristics on movement of bubble TCLs

We select several physical quantities related to bubble TCL movement and study the impact of surface parameters on them. Physical quantities include:

$$u^*_{\mathrm{TCL},n} = \frac{u_{\mathrm{TCL},n}}{u_{\max}} \tag{17}$$

$$u^*_{\mathrm{avg}} = \frac{\sum_{n=1}^{N} \int_0^{t_{\max}} \frac{u^*_{\mathrm{TCL},n}\,\mathrm{d}t}{t_{\max}}}{N} \tag{18}$$

$$a^*_{\mathrm{TCL},n} = \frac{u^2_{\mathrm{TCL},n}}{L_{\mathrm{ch}}} \tag{19}$$

$$a^*_{\mathrm{avg}} = \frac{\sum_{n=1}^{N} \int_0^{t_{\max}} \frac{a^*_{\mathrm{TCL},n}\,\mathrm{d}t}{t_{\max}}}{N} \tag{20}$$

$$Ca = \frac{\mu u_{\mathrm{TCL}}}{\sigma} \tag{21}$$

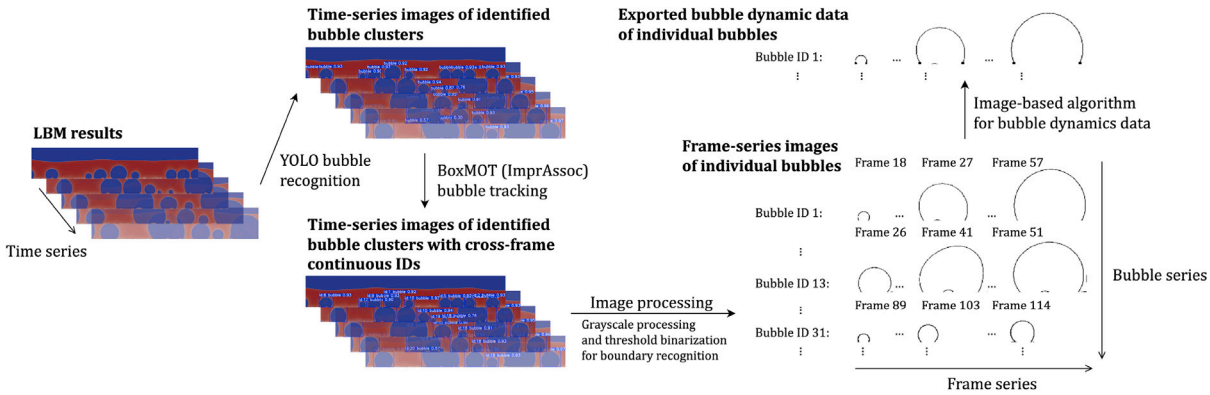

Fig. 14. Data mining process for contact line dynamics parameters.

where n denotes the bubble ID number, N denotes the total number of bubbles, $t_{\max}$ denotes the simulation time, $u_{\max}$ and L_{ch} are two characteristic parameters in our LBM simulation and respectively defined as the maximum speed 20 m/s and characteristic length 1.0×10^{-4} m. The subscript $_{\mathrm{avg}}$ represents the average *local* (for specific single bubble) or *global* (for bubble cluster) physical quantity over all bubbles. The non-dimensionalization of speed and acceleration can quantitatively describe the dynamic behavior of bubble TCLs. The capillary number Ca is also used to describe the TCL motion on the surface, where μ and σ are respectively the dynamic viscosity and surface tension of water determined based on thermal properties.

Fig. 15 shows the distribution of the average local dimensionless speeds of all TCLs over a period of more than 100,000 Δt under the conditions of $\Delta T^* = 0.35$, $\xi = 5.0$, and $Ra = 0$. Under the MOT framework, when two bubbles coalesce, the ID of the new bubble is inherited from one of the original two bubbles, thus these long-life-period bubbles exclude those merging within a relatively short time to avoid random errors. The bubble ID also indicates the order of bubble generation from the beginning to the end. In order to demonstrate the speed of each bubble, the average local TCL speed of each bubble is calculated for 0%~30%, 30%~70%, and 70%~100% of its existence period as $u^*_{0\sim30}$, $u^*_{30\sim70}$, and $u^*_{70\sim100}$, respectively, and the calculation method is the same as that of u^*_{global}. The speed here is always calculated using absolute values of velocities. The results show that u^* of bubbles generated in the early stages is generally higher than those generated later. Besides, for early-stage generated bubbles, the early speed ($u^*_{0\sim30}$) is higher than the late speed ($u^*_{30\sim70}$ and $u^*_{70\sim100}$). The surface space in the early stages of simulation is not yet occupied by other bubbles, allowing newly generated bubbles to grow freely and coalesce with other bubbles, which results in high values of $u^*_{0\sim30}$. As these bubbles grow, their u^* gradually slows down, which is reflected in the relatively lower $u^*_{30\sim70}$ and the decrease in the $u^*_{0\sim30}$ of subsequent bubbles.

Fig. 16 shows the variation of u^*_{global}, Ca and a^*_{global} under different ΔT^*. Further investigation is conducted on the average global speeds of advancing speed and receding speed of TCLs. Since our research object here is nucleated bubbles, we define the advancing speed as the speed of TCLs moving from vapor phase to liquid phase, and the receding speed as the speed of TCLs moving from liquid phase to vapor phase. The results indicate a high correlation between u^*_{global}, a^*_{global}, and ΔT^*. u^*_{global} rapidly increases after $\Delta T^* \geq 0.20$, reaching a plateau after $\Delta T^* \geq 0.30$ with a slowing growth trend. This growth trend is also reflected in the associated a^*_{global}. Similarly, a^*_{global} also experiences rapid growth after $\Delta T^* \geq 0.30$. As a function of u^*_{global}, Ca exhibits ΔT^*-related characteristics similar to u^*_{global}. As ΔT^* increases from 0.20 to 0.30, u^*_{global} and Ca increase by about 93%, while a^*_{global} increases by about 216%. This is consistent with the conclusion with TCL we obtained through void fraction in Section 3. Considering the relationship between advancing speed, receding speed, and average speed, one can find that the advancing speed is always greater than

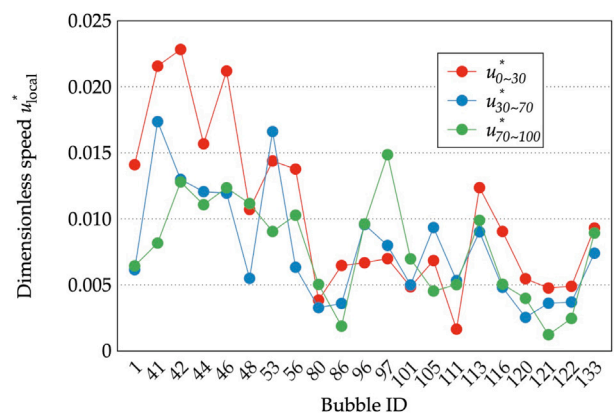

Fig. 15. Evolution of average dimensionless TCL speeds of bubbles over simulation time under the condition of $\Delta T^* = 0.35$ $\xi = 5.0$ and $Ra = 0$.

the receding speed, and the average speed is closer to the advancing speed. This is because our results mainly focus on the growth process of bubbles, in which case the advancing speed dominates relative to the receding speed. The few occurrences of receding speeds happen when bubbles come into contact with the upper boundary and rupture, as well as when neighboring bubbles of a ruptured bubble are forced to move towards voids. These two cases only account for a small part of the entire simulation time, making the statistically averaged global speed closer to those of advancing speed. Fig. 17 shows the distribution of u^*_{TCL} in the simulation cases of different ΔT^* as a supplement of Fig. 16. The result shows good consistency in the distribution of u^*_{TCL} in each case, and an increase in ΔT^* will lead to a rightward shift in overall dimensionless speeds.

Fig. 18 shows the influence of ξ on u^*_{global}, Ca and a^*_{global}. Overall, u^*_{global}, Ca and a^*_{global} show a decreasing trend as ξ increases. As ξ increases from 2.0 to 3.0, there is a decrease of approximately 43% in u^*_{global} and Ca, and a decrease of about 57% in a^*_{global}. When ξ further increases to 7.0, the change in u^*_{global} does not exceed 10%. The correlation between advancing speed, receding speed, and average speed is similar to Fig. 16, with average speed closer to advancing speed. This reflects that the hydrophilic surface makes it more difficult for TCLs to move from vapor phase to liquid phase, while the hydrophobic surface makes it easier (manifested as higher advancing and average speeds, and lower receding speeds). Larger values of ξ result in relatively smaller differences between advancing and receding speeds. As ξ increases from 3.0 to 7.0, the difference between the advancing speed and u^*_{global} decreases from 20% to 3%, and the difference in receding speed decreases from 40% to 5% (relative to average speed).

When analyzing the speed and acceleration of surface with Ra, the distance is multiplied by 2 for the structure depicted in Fig. 1 because

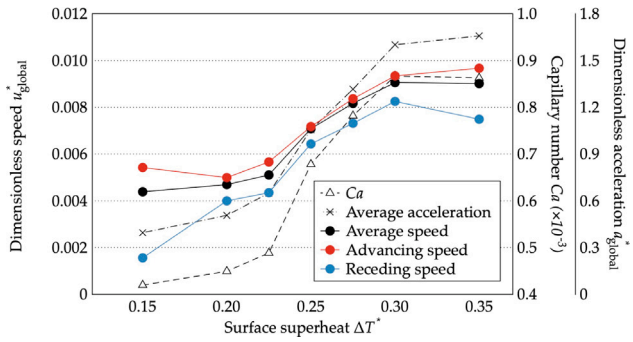

Fig. 16. Relationship between u^*_{TCL} (solid line, left vertical axis), Ca (dashed line, right vertical axis), and a^*_{TCL} (dashdotted line, right vertical axis) with ΔT^* under the condition of $\xi = 5.0$ and $Ra = 0$.

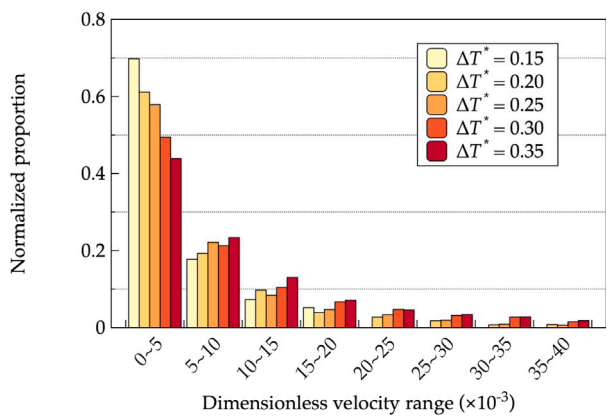

Fig. 17. Distribution of u^*_{TCL} with ΔT^* under the condition of $\xi = 5.0$ and $Ra = 0$.

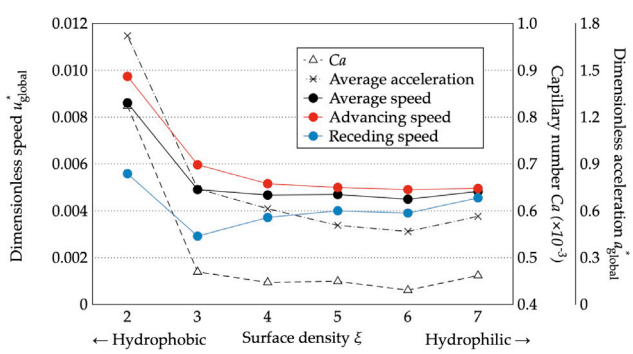

Fig. 18. Relationship between u^*_{TCL} (solid line, left vertical axis), Ca (dashed line, right vertical axis), and a^*_{TCL} (dashdotted line, right vertical axis) with ξ under the condition of $\Delta T^* = 0.20$ and $Ra = 0$.

the geometric structure increases the distance of TCL movement. Due to numerous cases, examples related to roughness are grouped and solved for average values. Fig. 19 shows the influence of Ra on u^*_{global} and a^*_{global}. The results indicate that compared to a smooth surface without roughness, a surface with roughness significantly increases u^*_{global} and Ca, with peak u^*_{global} around $Ra = 0.5\sim1.0$ μm. When $Ra \geq 2$ μm, u^*_{global} further increases with increasing Ra. The relationship between advancing speed, receding speed, and average speed remains consistent, showing good regularity. Also, a^*_{global} approximately increases as Ra increases. u^*_{global} are similar between $Ra = 0.5\sim1.0$ μm and $Ra = 8.0\sim10.0$ μm; however, a^*_{global} are higher within $Ra = 8.0\sim10.0$ μm. Based on observed bubble behavior, this is because when significant

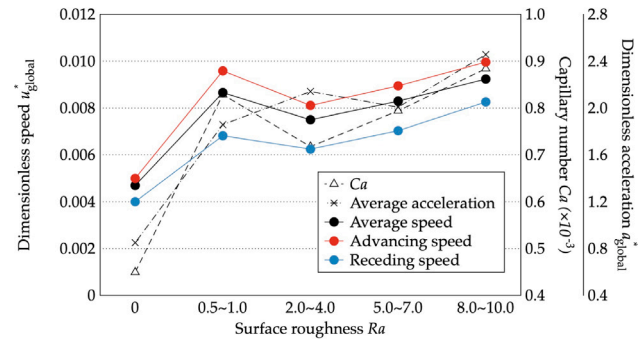

Fig. 19. Relationship between u^*_{TCL} (solid line, left vertical axis), Ca (dashed line, right vertical axis), and a^*_{TCL} (dashdotted line, right vertical axis) with Ra under the condition of $\Delta T^* = 0.20$ and $\xi = 5.0$.

rough structures are present, TCL tends to be *pinned* and *jump* between different structures rapidly (moving quickly from one edge of a structure to another, as shown in Fig. 8), resulting in higher accelerations being exhibited by higher levels of Ra.

Compared with the findings discussed in Section 3.2, it is observed that there is not a complete correspondence between TCL dynamic behavior and macroscopic heat transfer coefficients. For the dynamic laws dominated by ΔT^*, u^*_{global} shows a high correlation with ΔT^*, and the relatively high u^*_{TCL} at high temperatures may explain one of the reasons for the high heat flux q. In examples dominated by Ra variations, Ra near one of the peak of u^*_{global} can correspond to the peak of q, which to some extent explains the correlation between u^*_{global} and q. However, further increases in Ra do not result in an associated increase in q, despite further growth in u^*_{global}. We use the effect of TCL being pinned between structures to explain the negative impact of Ra's increase on q, and based on the observed jumping motion of TCLs at high Ra during actual bubble processes and the higher a^*_{global} to explain why there is still a high u^*_{global} in this case. The relationship between ξ and u^*_{global} exhibits a trend completely opposite to that observed with q. Under conditions of low ξ, although TCL moves swiftly and freely, processes such as bubble merging – caused by bubbles covering nucleation sites or the difficulty in detaching on surfaces with low wettability – impeded its motion and concurrently hinder macroscopic heat transfer. Conversely, an increase in ξ results in the advancing speed approaching the receding speed; thus, under conditions of high wettability, the receding side of TCL closely followed the advancing side. This behavior could also explain the higher values of q observed.

Another parameter worth discussing is the apparent bubble contact angle θ^*. The general definition of the contact angle θ of a liquid on a surface is described by the angle formed by a static liquid droplet on the surface, as shown in Figs. 1 and 2. Here we discuss the apparent bubble contact angle θ^* between bubbles and the surface. We calculate θ^* in the same way as Eq. (12), except that l_d and h_d are used to refer to the contact length between bubbles and the surface and the height of bubbles, respectively. l_d is determined by two boundary points obtained by the post-processing method, and h_d is determined by the height of the bubble cropping box. The θ^* is the angle formed by the boundary between the liquid phase and the vapor phase, and the boundary between the vapor phase and the solid phase of TCLs, and is thus the supplementary angle of the contact angle θ in the general sense. The LBM results are dynamic processes of bubble nucleation and coalescence, and the captured bubbles are not always approximated to a liquid droplet shape that is stationary on the surface. However, in our simulations, most bubbles are in the process of nucleation from the surface, and we calculate the average bubble contact angles for the entire simulation process. Therefore, although there is some randomness in bubble behavior and this method might not always provide an accurate bubble contact angle, it is still worth analyzing its

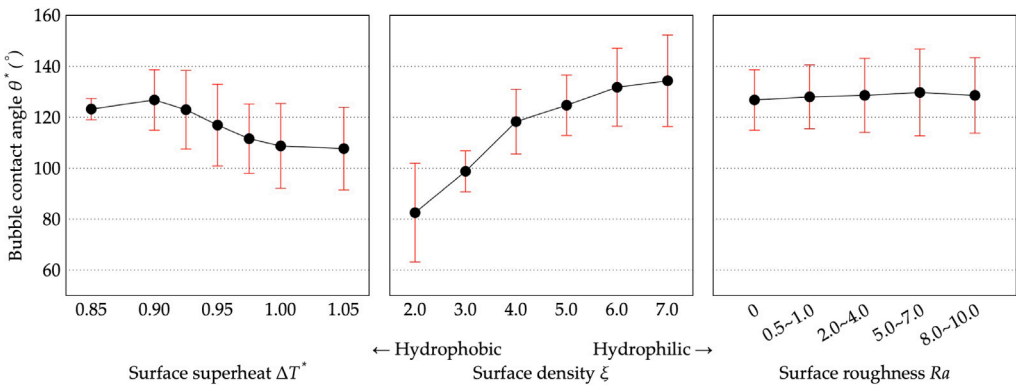

Fig. 20. Relationship between θ^* with ΔT^*, ξ, and Ra.

regularity from a statistical perspective. Fig. 20 provides the relationship between the bubble contact angle θ^* of all bubbles across frames in a single case and three surface parameters ΔT^*, ξ, and Ra. Error bars are annotated using the standard deviation of the data to indicate variability. The results indicate that θ^* is mainly controlled by ξ. As the surface wettability transition from hydrophobic to hydrophilic, θ^* increases. The characteristics of θ^* are opposite to the contact angle of the droplets shown in Fig. 2. ΔT^* also shows a certain influence on θ^*, with θ^* decreasing as ΔT^* rise, but the effect of ΔT^* on θ^* is not as obvious as that of ξ. A special case occurs when $\Delta T^* = 0.15$, which may be due to the small number of bubbles in this particular scenario. There is no apparent regularity between Ra and θ^*.

5. Conclusions

In this paper, two-dimensional numerical simulation studies of surface wetting properties and pool boiling on a flat surface were conducted using LBM. The pseudopotential method was added to solve two-phase flow problems, while the DDF method was added to solve thermal problems. The correlations of bubble nucleation and heat transfer behavior with surface wettability, surface roughness and density were obtained. Conclusions related to the bubble nucleation, growth and departure behavior, and heat transfer have been obtained as follows:

1. Surface density is a comprehensive expression of surface wettability and wetting mode. Increasing the surface density decreases the contact angle and causes a change in the wetting mode. The introduction of roughness causes changes in the nonlinear relationship between contact angle and surface density.
2. LBM provides useful descriptions of the processes of bubble nucleation, coalescence, rupture, and re-nucleation. The surface superheat significantly intensifies the nucleation and heat transfer, which is also the main factor in the peaking of heat transfer coefficient. Increasing the surface superheat by 0.05 from 0.20 results in a 30% increase in void fraction and also leads to a slight rise in heat flux. The introduction of roughness has a complex impact on surface wetting and heat transfer. Roughness enhances heat transfer within a specific range but becomes less efficient beyond. Introducing roughness increases heat flux by up to 10% when $Ra = 1.0$, but decreases it by up to 25% when $Ra = 10.0$ compared to a smooth surface.
3. For different roughness, the effect of surface density on void fraction and heat transfer is similar. The void fraction decreases as surface density increases, indicating the transition from hydrophobic to hydrophilic. The heat flux increases with increasing surface density but stabilizes within a 10% range after surface density exceeds a specific threshold value, in line with the effect of wetting modes on heat transfer.
4. The post-processing method, which combines ML and computer vision, is employed to export the global contact line dynamics data from the LBM results. The YOLO model and the MOT method are utilized to achieve the most accurate contact line dynamics data. The results reveal that surface superheat, surface density, and surface roughness each exert distinct impacts on the global average contact line dynamics speed and acceleration. Increasing the surface superheat from 0.20 to 0.30 results in a significant 93% increase in both the global average bubble speed and the capillary number. Conversely, the increase in surface density from 2.0 to 7.0 results in a reduction of the global average bubble speed and the capillary number. Significant reduction occurs between 2.0 and 3.0, resulting in approximately a 43% decrease in the global average bubble speed and the capillary number, while the subsequent changes are limited to a small range. The increase in surface density also causes the advancing and receding speeds to converge towards the average speed. The initial peak in global speed correlates with the roughness that facilitates a high heat transfer flux. The influence of each surface parameter on acceleration is consistent with its effects on speed. The bubble contact angle is mainly controlled by surface density, but surface superheat also has a certain influence.

In the LBM section, we clearly described the effects of three surface parameters on bubble behavior and heat transfer behavior by introducing void fraction and heat flux. This part of the conclusions focused on the nucleation and coalescence behavior of bubble clusters near the surface in a small and confined area. It provided insights into bubble and heat transfer behaviors in pool boiling when combined with studies on growth and detachment processes of individual bubbles in other CFD methods. The ML method was completely separated from the CFD section, achieving statistical descriptions of dynamic parameters for complex large numbers of bubble clusters through efficient identification and derivation of contact line dynamics parameters. This description is not only applicable to the scenario in this work but may also be extended to other works. Compared with other work, this work efficiently realized the description of bubble cluster generation and boiling heat transfer behavior in a very small and confined domain using LBM, and further coupled the ML method to achieve statistical significance in describing the dynamic parameters of bubble clusters throughout the entire boiling process. The LBM method is ideal for accurately describing the nucleation behavior of bubble clusters in small and confined spaces. It aids in understanding and identifying the dynamics of bubble clusters through ML methods, offering valuable statistical insights. Based on these results, we attempted to establish a connection between macroscopic heat transfer behavior and microscopic bubble dynamics parameters. We summarize in Table 2 all surface parameters studied in this work for their effects on heat transfer and bubble dynamics behavior.

Table 2
Effects of surface parameters on heat transfer and bubble dynamics behavior.

Surface parameters (with the increase of)	Range	Effects on			
		Void fraction ξ	Heat flux q	Global speed u^*_{TCL} and Ca	Bubble contact angle θ^*
Surface superheat ΔT^*	0.20~0.30	Increase	Increase	Increase	Slightly decrease
Surface density ξ	3.0~7.0	Decrease	Increase and then stabilize	Decrease and then stabilize	Increase
Roughness Ra	0~10.0 μm	Decrease and then increase	Increase and then decrease	Increase relative to smooth surface	No obvious correlation

Considering the difficulty of calculating the pool boiling process on a flat surface based on hybrid thermal LBM, this study used a two-dimensional LBM calculation with the BGK operator. We aim to utilize ML methods to analyze the statistical properties of bubble contact line dynamics data from two-dimensional pool boiling simulations, which can offer insights into related challenges. However, it is important to note that factors such as bubble coalescence in multiple directions make three-dimensional scenarios more complex. Utilizing advanced LBM like the multiple-relaxation-time (MRT) operator for three-dimensional thermal simulations is a promising research approach. We are also currently applying our data mining process to pool boiling experimental data in order to collect bubble nucleation information, with a specific focus on the subsequent bubble departure process. By examining additional bubble dynamic parameters like the position of nucleation points and growth rate, a clearer understanding of the bubble nucleation process is expected to be achieved.

CRediT authorship contribution statement

Jiajun Wang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Akiko Kaneko:** Writing – review & editing, Supervision, Resources. **Biao Shen:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2025.129000.

Data availability

Data will be made available on request.

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